

Cataloger Judgment and Human-in-the-Loop: Artificial Intelligence in Library Technical Services

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Abstract

Artificial intelligence is increasingly woven into everyday life and has become difficult to avoid or ignore. In library technical services, however, its value depends on the work it is asked to do, the evidence available for review, and the professional judgment surrounding its use. Building on this context, this chapter examines recent literature on AI in metadata creation, cataloging support, quality control, with attention to both practical applications and ongoing limitations. Furthermore, it presents local exploratory work involving a Catalog Buddy developed through iterative MARC review and a rule-guided quality-control project for historical photograph metadata. In both projects, documented methods were used to keep source evidence, local practice, and cataloger decisions visible. As a result, AI proved most useful for structured comparison, recurring patterns, and triage, while it was less dependable when the work required item-specific context, authority research, subject analysis, classification, or knowledge of a local collection. Importantly, neither project supports autonomous cataloging. Instead, the findings demonstrate how carefully scoped tools can assist review while responsibility remains with trained staff. Finally, questions of bias, privacy, transparency, labor, and public description are treated as part of the workflow itself and not as concerns to be considered only after adoption.

Keywords

Artificial Intelligence; Technical Services; Cataloging; Metadata Quality; Human-in-the-Loop; Responsible AI

1. Introduction

1.1 A charged debate in academic libraries

Debates about artificial intelligence in academic libraries can become intensely charged. Some librarians see an opportunity to expand critical literacy instruction, test new service models, and help students understand the tools they already use. Others are more wary. They point to privacy, environmental cost, bias, dependence on systems that are difficult to examine, and the possibility that convenience may displace careful thought. These concerns reflect different understandings of what the library owes its users, its institution, and the public. Bridges et al. (2024) argue that trust should shape the pace of adoption and that collaboration can help

libraries protect access, privacy, and academic freedom while remaining part of wider conversations about AI. Local decisions about teaching, licensing, restricting, or piloting these tools are therefore as much about institutional values as technical capacity.

In cataloging and technical services, discussions typically focus on the practical aspects of the work. Staff are primarily interested in whether AI can assist with repetitive or time-consuming review tasks, rather than its potential as a general-purpose chatbot. Desired applications include identifying inconsistencies, screening metadata batches, detecting possible duplicates, and flagging names requiring authority work. These tasks, however, depend on contextual knowledge, established standards, local practices, and professional judgment. The central question is whether AI can streamline review processes while ensuring that interpretation and final decisions remain with library staff. The Program for Cooperative Cataloging Task Group on Strategic Planning for AI and Machine Learning (2024), hereafter the PCC Task Group, similarly emphasizes the importance of accuracy, accountability, and maintaining control over library description.

1.2 A profession accustomed to technological change

Libraries have previously navigated significant technological transformations. Developments such as online catalogs, integrated library systems, discovery layers, electronic resources, and digital repositories have each reshaped library operations, yet none have eliminated the necessity for skilled staff or professional judgment. Yoon et al. (2022) note that libraries have consistently adapted to emerging technologies, even amid uncertainty regarding their long-term implications. Similarly, Sonawane et al. (2024) characterize university libraries as dynamic, technology-rich environments rather than passive repositories. Artificial intelligence may similarly influence the division of labor, although its full impact remains uncertain. While staff may not interact directly with generative systems daily, they will increasingly need to identify where AI is embedded in their tools, understand the data these systems utilize, and determine when outputs require additional scrutiny.

This historical context is significant because technological advancements do not diminish the profession's commitment to service. Ranganathan's five laws continue to provide a valuable framework for considering access, the relationship between readers and resources, the efficient use of readers' time, and the library's potential for growth. Kalbande et al. (2025) apply these principles to artificial intelligence by evaluating whether new tools genuinely enhance resource discoverability and usability, and whether they support library adaptation without detracting from its fundamental mission.

1.3 Cautious adoption and an unsettled evidence base

Adoption of artificial intelligence in cataloging and metadata remains limited. In its 2024 environmental scan of 193 practitioners, the PCC Task Group found that only 11.5 percent were using AI to automate routine technical services tasks. Respondents who had experimented with AI reported inconsistent outcomes and expressed concerns regarding hallucination, bias, copyright, privacy, security, labor equity, and environmental impact (PCC Task Group 2024). The survey characterizes the profession as proceeding cautiously rather than rapidly embracing automation.

The published literature is also more cautious than promotional accounts of AI sometimes suggest. Moulaison-Sandy and Coble (2024) find that AI can assist with selected cataloging tasks when appropriate oversight is built into the work. Engel et al. (2025) offer a more guarded assessment, finding little evidence that removing human review produces meaningful efficiency gains without reducing quality. In practice, the key issue is the time and expertise required to make the generated output usable. A system may produce an answer quickly while leaving staff with another layer of review.

This chapter has two purposes. First, it synthesizes current literature on AI in metadata creation, cataloging support, quality control, ethical implementation, and staff preparation in library technical services. Second, it examines two exploratory local case studies—a Catalog Buddy developed through iterative MARC review and a rule-guided quality-control project for historical photograph metadata—to show how human-in-the-loop workflows can make evidence, local practice, and cataloger decisions visible.

2. Defining AI in Technical Services

2.1 Technical services as knowledge infrastructure

Technical services work provides the descriptive and systems foundation that makes a collection discoverable and usable. A catalog record or call number may look straightforward, but it reflects decisions made across many areas of the library. When metadata is incomplete, inconsistent, or too limited, valuable materials can remain effectively hidden, reducing the value of the library's investment in acquiring, preserving, and digitizing them (Ayccock 2025). Seen in this way, technical services are not simply behind-the-scenes processing. It is part of the library's knowledge infrastructure.

This work extends beyond the mere application of established rules. Staff members interpret standards, develop local practices, and adapt frameworks such as Machine-Readable Cataloging (MARC) and the Bibliographic Framework (BIBFRAME) to address the specific needs of collections and systems. Consequently, technical-services work is both structured and interpretive, guided by standards yet shaped by contextual factors.

MARC 21 and BIBFRAME are maintained by the Library of Congress and provide key standards infrastructure for this discussion (Library of Congress n.d.-a, n.d.-c).

These challenges frequently extend beyond the boundaries of a single department. Affum and Dwomoh (2023) highlight the importance of collaboration among librarians, researchers, and AI specialists. Such partnerships are most effective when domain knowledge informs problem definition and evaluation from the outset, rather than being incorporated after system design is complete.

2.2 Terms, tools, and scope

For this chapter, AI is a broad term for tools that recognize patterns, generate or transform language, compare information, and support review. What matters most is the authority assigned to a tool. It may help staff notice or organize evidence, but it does not become the final source of a cataloging decision.

The primary challenge is practical. Technical services departments manage metadata originating from vendor records, digital collections, mixed formats, and longstanding backlogs. Metadata functions as more than a simple label; it provides the structure necessary for discovery, preserves provenance, and documents relationships among resources and entities (Greenberg et al. 2026). An AI tool is justified in this context only if it demonstrably improves a clearly defined task without obscuring uncertainty, compromising standards, or shifting the review burden elsewhere.

As table 1.1 shows, several MARC fields recur throughout the chapter examples and quality-control discussion. The table provides a plain-language reference for readers who are not catalogers.

Table 1.1. Common MARC fields referenced in this chapter

MARC field	Plain-language identifier	Use in this chapter
008	Fixed-length coded data field	Encodes dates, language, place, illustrations, bibliography/index evidence, and other coded information.
043	Geographic area code field	Connects coded geography with the geographic coverage reflected in subject access.
090	Local call number field	Used for locally assigned call numbers; may reflect local shelving practice.
245	Title statement field	Records title, subtitle, statement of responsibility, indicators, and transcription review.
246	Varying title field	Records alternate title forms users might search.
260	AACR2 publication field	Records publication information in older records; not automatically wrong in legacy records.
264	Production/publication/copyright statement field	Records production, publication, distribution, manufacture, and copyright information in RDA records.
300	Physical description field	Records extent, illustrations, maps, leaves/pages, dimensions, and item-in-hand description.
33X	Content/media/carrier fields	RDA fields that describe content type, media type, and carrier type.
500	General note field	Used for source of title, item format, limited-edition information, and contextual notes.
504	Bibliography/index note field	Used when the item contains bibliographical references or an index.

MARC field	Plain-language identifier	Use in this chapter
505	Contents note field	Used for contents, chapters, parts, or papers.
520	Summary note field	Used for neutral description of content and scope.
6XX	Subject access fields	Used for personal, corporate, topical, geographic, and genre/form subject access.
700/710	Added entry fields	Used for additional personal and corporate contributors.

2.3 Human-in-the-loop review and AI literacy

A human-in-the-loop workflow is not a final rubber stamp on automated output. In technical services, it means that a cataloger or metadata professional checks the source evidence, compares the output with standards and local practice, and decides whether a result should be accepted, revised, rejected, or documented for later review. The system may organize evidence or surface a pattern, but it cannot assume responsibility for the public record.

AI literacy supports this review. Staff need enough understanding of how AI tools produce output to recognize fluent but unsupported claims, privacy risks, bias, and uncertainty. In this chapter, AI literacy is treated as practical professional judgment: knowing when a tool is useful, when it requires additional checking, and when it should not be used. Harisanty et al. (2024) place these abilities within broader competencies such as critical thinking, adaptability, communication, creativity, and leadership.

2.4 Cataloger judgment and accountability

Cataloger judgment is the professional reasoning that turns evidence into reliable metadata. It informs decisions about transcription, supplied information, access points, subject representation, standards, local policy, and whether a machine-generated suggestion is useful, incomplete, biased, or wrong.

This judgment is most visible when evidence is incomplete or contradictory. Resources may present conflicting dates, uncertain authorship, variant titles, ambiguous publication information, or inherited records that describe a related edition rather than the item in hand. Standards such as RDA, MARC 21, LCSH, Dublin Core, MODS, and local guidelines create structure, but their application still requires interpretation.

The official standards and vocabularies cited here include RDA Toolkit, MARC 21, LCSH, Dublin Core, and MODS (Dublin Core Metadata Initiative 2006; Library of Congress n.d.-b, n.d.-c, n.d.-d; RDA Toolkit 2026).

An effective quality-control workflow must therefore distinguish among genuine errors, optional normalization, accepted local practice, and issues that require cataloger discretion. A blank **090 local call number field**, an older **260 AACR2 publication field**, a retained vendor field, or

missing punctuation may or may not require correction depending on the record history, review purpose, and local remediation goals.

As figure 1.1 shows, the decision categories used in this chapter keep AI-assisted review grounded in professional judgment. The categories show that not every machine-identified issue should be treated as a production error; some findings require high-risk review, some are optional normalizations, some reflect local practice, and some records should be accepted as they are.

Figures 1.1 Categories for Review: failure or success replied by Cataloging Buddy

Categories for AI-Assisted Cataloging Quality control



Accountability remains with the cataloger. Before accepting a suggestion, the reviewer must determine whether it is supported by evidence, follows the relevant standard, aligns with local practice, can be defended ethically, and improves discovery. AI can assist with comparison, drafting, and triage, but it cannot answer for the record.

3. Metadata Creation and Enhancement

3.1 AI-generated metadata as draft metadata

AI-generated text should be regarded as a preliminary draft. It can assist catalogers in initiating descriptions, identifying potential topics, or standardizing preliminary notes. However, linguistic fluency does not guarantee accuracy. Reviewers must assess whether the text is accurate,

specific, contextually appropriate, sensitive to subject matter, and consistent with authorized terminology and local practices.

The level of review required should correspond to the degree of interpretation involved. Extracting a visible date from a title page is a straightforward task, whereas determining a work's primary subject requires more nuanced judgment. Similarly, standardizing repeated labels presents less risk than establishing relationships between entities. While AI can assist in each scenario, the evidentiary basis and potential consequences differ significantly.

3.2 Summaries, notes, and remediation

One practical application of AI is drafting a MARC **520 summary note** or repository abstract. Catalogers may supply an introduction, abstract, table of contents, or item description and request a concise, neutral summary. For lengthy or interdisciplinary sources, this approach can provide a valuable starting point for further refinement.

Among cataloging and metadata professionals who had used AI, translation and summarization were the most commonly reported applications (Chen and Li 2024). These tasks are a reasonable starting point because the source can be supplied directly and the generated text can be checked against it.

Draft summaries require careful comparison with the original source. Summaries must accurately reflect the scope and emphasis of the work, avoiding promotional language or unsupported claims. They should be comprehensible to readers beyond a specialized audience. Stylistic refinement does not constitute improvement if it alters the intended meaning.

Language models can also facilitate remediation by identifying repeated descriptions, standardizing note phrasing, or segmenting unstructured text into fields compatible with Dublin Core, Metadata Object Description Schema (MODS), or local schemas. Their primary value is in efficiently processing repetitive structures. However, these models lack the contextual understanding possessed by collection specialists.

Dublin Core and MODS are included here as examples of metadata schemas that can structure descriptive data outside or alongside MARC environments (Dublin Core Metadata Initiative 2006; Library of Congress n.d.-d).

Efficiency claims must be evaluated in relation to the descriptive requirements of each collection. Corrado (2021) emphasizes the importance of balance: metadata should adequately support discovery, access, and use, but excessive processing may yield diminishing returns. While AI can improve efficiency for repetitive tasks, it cannot independently determine the optimal level of description for specific collections or audiences.

Tian's (2026) study of a born-digital legal collection shows both the promise and the limits of this approach. Generative AI reduced the time needed for structured tasks such as title extraction, standardized abstract creation, and file organization. Work that had taken one librarian and two staff members two to three days was completed by one person in less than half a day. Keyword matching was far less dependable, however, because semantic differences and false matches required extensive professional review. The collection was also small and consistently

structured, so the findings should be applied cautiously to collections with more uneven or complicated metadata.

3.3 Subject analysis and controlled vocabularies

Subject analysis demonstrates that word recognition does not equate to understanding a resource's content. AI systems can extract potential topics from titles, abstracts, tables of contents, or summaries and propose topical, geographic, or genre/form terms. This functionality may assist catalogers in unfamiliar subject areas, but it does not establish which concepts warrant inclusion as access points in the record.

Determining aboutness relies on identifying the work's primary emphasis. A topic addressed in a single chapter may not reflect the overall focus, and a historical study may reference specific places or groups without centering them in its argument. Suggested terms may be overly broad, excessively narrow, outdated, or influenced by bias. Catalogers must ultimately determine the principal subject of the resource and the appropriate level of specificity for its expression.

Daniels (2026) describes this difficulty from a cataloger's perspective. In her testing, AI suggestions varied in both directions, sometimes offering terms that were broader and sometimes narrower than the level she considered appropriate. Her findings return to a familiar cataloging principle: a heading should be as specific as the resource supports, not merely tied to language in the text.

Controlled vocabularies such as the Library of Congress Subject Headings (LCSH) and Faceted Application of Subject Terminology (FAST) ensure consistency in subject access. AI-generated phrases may resemble authorized headings but may not conform to established standards, potentially misusing subdivisions, qualifiers, or geographic forms. In performance evaluations with ChatGPT, Gemini, and Copilot, Dobreski and Hastings (2025) determined that none of these systems was adequate for assigning Dewey Decimal Classification (DDC), Library of Congress Classification (LCC), or LCSH. Conversely, Suominen et al. (2022) demonstrated that tools like Annif can be effective when suggestions are grounded in controlled vocabularies, local training data, and human oversight.

Official documentation for LCSH and FAST is especially important because AI-generated phrases may resemble authorized headings without actually matching a controlled vocabulary (Library of Congress n.d.-b; OCLC n.d.).

These published findings align with local exploratory observations from Copilot and ChatGPT testing, which produced plausible but inconsistent subject suggestions. Catalog Buddy, a locally developed GPT-based cataloging quality-control assistant configured through iterative MARC review and cataloger corrections, is discussed in Sections 4.7–4.9. It was designed to organize review, surface candidate issues, and document cataloger decisions rather than assign final headings or produce autonomous records.

In local testing, the most useful subject-related output was not a production-ready heading but a set of candidate terms, rationales, and review questions that could guide authority checking and item-level evaluation. Unauthorized phrases could still serve as search leads, but they were not

appropriate for production metadata until the correct form and relevance to the resource were verified.

Authorization constitutes only one aspect of the review process. Controlled vocabularies reflect historical structures and may contain contested language, which AI systems can inadvertently perpetuate or may fail to recognize terms preferred by communities represented in local collections. Ethical considerations should be integrated into subject quality control rather than treated as an isolated, subsequent step.

Reviewers should assess whether a heading accurately represents the resource, avoids overemphasizing secondary topics, and employs language that is neither harmful nor misleading in context. Certain decisions may necessitate consultation with community stakeholders or adherence to an institution's inclusive description policy.

3.4 Semantic enrichment and digital collections

Semantic enrichment establishes connections rather than simply increasing textual content. Associating names with authority records, subjects with external vocabularies, or creators with identifiers such as Open Researcher and Contributor ID (ORCID) or Virtual International Authority File (VIAF) enhances interoperability and facilitates user navigation across systems. Greenberg et al. (2026) situate this activity within a broader metadata ecosystem, where data structure, value, syntax, and encoding standards ensure information is findable, traceable, interoperable, and reusable. For AI, high-quality metadata serves as both output and infrastructure, enabling entity discovery, provenance tracking, and retrieval constraints. The utility of any link ultimately depends on the accuracy of the asserted identity and relationship.

Digital collections present distinct opportunities. Optical character recognition (OCR) and handwriting recognition generate searchable text, while named-entity recognition identifies people, places, organizations, and dates. Computer vision can suggest visible objects or scenes. These technologies can broaden access and facilitate large-scale description, yet recent photographic projects demonstrate the significant quality control required for multimodal workflows (Facklam et al. 2025; Wolfe 2026).

Recent photographic projects highlight both the potential and the extensive review required. Facklam et al. (2025) assessed a multimodal model using model selection, batch processing, and evaluation for completeness, accuracy, consistency, and bias. Wolfe (2026) implemented a vision-enabled workflow for 2,263 photographs, finding that generated subject terms corresponded to FAST in 64 percent of cases. This project also incorporated transparency notices and integrated local quality control into the descriptive workflow.

Extraction tools are prone to characteristic errors. Image recognition may misidentify individuals, objects, locations, or cultural contexts. OCR inaccuracies can impair search functionality, and entity extraction may conflate persons with places or merge similar names. Contextual information essential for historical photographs and special collections is often absent from the image or text alone.

Local and historical collections are particularly resistant to generic descriptions. AI models may identify visible elements but overlook their significance, relevance to local history, or the

potential impact of labels on represented communities. Human expertise is essential to provide this contextual understanding.

In practice, digital-collections workflows require multiple layers of review. Automated tools may generate searchable text, entity candidates, object labels, or brief descriptions. Staff subsequently consult the image, collection documentation, donor restrictions, local history, community expertise, and inclusive description policies. While technology can expand descriptive coverage, human input provides the contextual meaning necessary for accurate representation.

4. Cataloging Support

4.1 Task-specific MARC record review

The structure of MARC facilitates the definition of certain machine-assisted tasks. Moulaison-Sandy and Coble (2024) contend that artificial intelligence can support specific aspects of book cataloging when tasks are narrowly defined and oversight is incorporated. For example, a tool may verify 245 title statement field indicators, identify missing subfields, validate 264 production/publication/copyright statement field date logic, detect conflicts between the 300 physical description field and 33X content/media/carrier fields, assess fixed-field coding, or highlight discrepancies between record versions. However, system configuration is critical. A generic checker may misinterpret local fields, legacy practices, or artifacts introduced by platforms such as Connexion or Alma.

The workflow must explicitly define the system's permitted actions. The system may flag specific fields, explain the relevant cataloging rule, suggest corrections, or direct attention to authority or item-in-hand verification. Ultimately, the cataloger determines whether a finding constitutes an error, optional cleanup, or an accepted local practice.

Dividing the cataloging process into discrete steps generally yields more effective results than requesting a complete record at once. Tasks such as data extraction, date validation, note drafting, access-point suggestions, and final quality control (QC) can be evaluated independently, with the level of review adjusted according to the risk associated with each step.

Aycock's (2025) evaluation of CatalogerGPT on nearly 100 e-book conference titles demonstrates the limitations of current systems. While the tool could generate descriptive fields and contents notes, it also mistranscribed titles and ISBNs, created malformed subfields, omitted essential data, and fabricated paper titles. Despite over 25 prompt revisions, each record still required professional review, and in some cases, manual cataloging proved more efficient.

4.2 Authority control and name disambiguation

In authority work, machine learning is most useful when it ranks candidates and preserves the evidence supporting those rankings. Dates, affiliations, co-creators, subjects, identifiers, language, and publication history can narrow a large set of possibilities, but a high score is not proof of identity. The record should retain the source evidence, the features used, the confidence level, and the cataloger decision.

False matches carry particular risk because they can attach works to the wrong person, corporate body, or subject entity and then travel through shared records or linked-data systems. AI-assisted authority work should therefore be treated as candidate generation and evidence organization rather than final identity resolution.

4.3 Classification and call-number support

Classification support is most useful at the beginning of research, especially for interdisciplinary materials. When provided with a title, contents, abstract, or subject description, a system can suggest plausible Library of Congress Classification (LCC) ranges and explain which concepts led to those recommendations.

The difficult part is choosing the work's dominant emphasis. A book on South Carolina textile mills and child labor could fit industrial, labor, social welfare, local, childhood, or economic history depending on its argument. The cataloger must consult the schedules, weigh the text as a whole, and account for local shelving practice.

Local publications make this boundary clear. The Catalog Buddy test involving a Clemson University official publication required decisions about official status, seriality, the local Cutter pattern, and the relationship between call number and subject headings. Topical similarity alone could not settle those decisions; the experiment is discussed in Sections 4.7–4.9.

Predictive classification is best used to identify starting points. The cataloger still chooses the schedule, applies tables and Cutters, and makes sure that classification and subject access tell the same intellectual story.

4.4 Bibliographic interpretation and linked-data workflows

Automated summaries are most reliable when generated from well-defined sources, such as an abstract, an introduction, a table of contents, or the complete text. Instructions should clearly specify the desired length, require a neutral tone, and prohibit unsupported assertions or promotional language.

The risk of error increases when the source material is incomplete or presents cultural and technical complexities. Reviewers should compare the generated summary with the original resource, assess the proportional representation of topics, and eliminate both interpretive overreach and any borrowed marketing language.

MARC continues to serve as the primary production format for many libraries, whereas BIBFRAME and other linked-data models focus on entities and relationships. AI-assisted tools can help identify candidate works, agents, subjects, and relationship types, or flag uncertain MARC-to-BIBFRAME mappings. However, these tools should not make modeling decisions without explicit documentation.

The Library of Congress frames BIBFRAME as a linked-data foundation for bibliographic description and as part of the transition path from MARC 21 (Library of Congress n.d.-a).

Maintaining provenance is critical during any data conversion process. Libraries should retain the source record, the transformation rule or model, the conversion date, and the identity of the

reviewer for each generated or transformed relationship. Without this documentation, errors may propagate and become increasingly difficult to resolve.

4.5 Supporting new catalogers and workflow learning

New catalogers may utilize AI tools as study aids to compare fields such as the 245 title statement field and the 246 varying title field, explain the functions of the 264 production/publication/copyright statement field, or generate self-assessment questions. These applications can make initial training less daunting and assist trainees in developing more effective inquiries.

AI-generated explanations cannot substitute for authoritative documentation or mentorship. Such explanations may be incomplete, inaccurate, or fail to account for local practices. Nevertheless, under appropriate supervision, they can supplement training and support new catalogers in formulating more precise questions.

Experiences with chatbot projects underscore the importance of limiting the scope of training tools. Cher (2025) observes that conversational systems may provide factually incorrect responses and perpetuate social biases. Consequently, a cataloging assistant should direct users to authoritative documentation, clearly indicate uncertainty, and refrain from presenting generated explanations as definitive rules.

Iacono (2025c) rejects the notion of AI as a thinking companion, instead characterizing it as a research tool whose value is contingent upon the user's judgment. This perspective aligns with supervised learning models, in which the tool retrieves, compares, and rephrases information, while the trainee and mentor retain responsibility for interpreting standards and evaluating evidence.

Departments should ensure that trainees have opportunities to construct records based on primary evidence. Exclusive reliance on correcting machine-generated output may result in familiarity with tool-specific patterns rather than foundational cataloging skills. Activities such as original description, authority research, classification, and written justification remain essential for comprehensive training.

4.6 Shared approach to the two local projects

Both local projects originated from a common practical question but were conducted separately due to their distinct objectives. The Catalog Buddy project focuses on MARC bibliographic review and record development, while the Byrnes project employs a rule-guided screening process for metadata describing historical photographs. A MARC assistant must reason across multiple fields, standards, and local practices, whereas a collection-level quality-control assistant performs a more limited function by sorting repeated patterns without asserting that spreadsheet checks can ensure complete descriptive accuracy.

This work was conducted as practitioner-led, design-based case-study research. The approach is exploratory rather than a randomized or controlled experiment. Cases were drawn from ongoing technical-services activities, model outputs were considered provisional, and cataloger corrections informed iterative revisions to the specifications. This methodology aligns with

recommendations for bounded tasks, expert oversight, deliberate pilot projects, and thorough documentation (PCC Task Group 2024; Proffitt et al. 2024).

The projects were guided by five questions:

1. Which structural or consistency checks are useful enough to support professional review?
2. Which findings still depend on item-in-hand evidence, authority work, local policy, visual inspection, or interpretation?
3. Can the assistant separate clear errors from review questions, optional normalization, local practice, and acceptable records?
4. How can cataloger corrections become reusable specifications, output categories, and later test cases?
5. What can be reported without overstating a development sample, a narrow QC screen, or an unfinished post-correction audit?

Both projects followed a common cycle: task definition, evidence identification, machine review, cataloger decision-making, outcome logging, specification revision, and subsequent testing. The cycle is described here rather than carried by a separate figure so that the local-project methodology remains clear in the text.

Human judgment serves as the pivotal element in both projects. The assistant generates candidate findings or statuses, while the cataloger determines their validity and the appropriate subsequent actions. Logs document the input, applicable rules or evidence, machine output, cataloger's assessment, corrections, and any modifications to the specification. Instances of disagreement are treated as valuable research evidence rather than dismissed as prompt failures.

4.7 Catalog Buddy: purpose and development corpus

The Catalog Buddy project tests whether a GPT-based assistant can act as a structured second reader for MARC bibliographic work. It was never designed to generate autonomous records. Its purpose is to help a cataloger notice structural conflicts, organize a review, compare fields, surface questions, and document why a proposed change is accepted, revised, rejected, or treated as local practice.

The development corpus comprises 20 records selected from the active workflow of a professional cataloger in an academic library and Special Collections context. This purposive sample includes government publications cataloged as books, older AACR2 monographs, RDA vendor stubs, original institutional and local-history records, a memoir, edited volumes, and an unpublished typescript. Two substantially correct legacy records served as low-error controls, enabling assessment of the assistant's ability to avoid unnecessary interventions.

Following the establishment of the 20-record corpus, *Ebb and flow, November 1950–July 1951*, was included as an extended-evidence case. This case tested the applicability of rules developed during the corpus—specifically RDA consistency, item-in-hand dates, government-body access points, series numbering, and alignment among the 300 physical description field, the 008 fixed-length coded data field, and the 504 bibliography/index note field—to a new

government publication, without incorporating the record into the development sample. The project remains ongoing.

4.8 Catalog Buddy: procedure and preliminary findings

Each case advanced through five iterative stages. First, the cataloger provided the MARC record along with any relevant item-in-hand or workflow context. Second, the Catalog Buddy generated a structured review, categorizing must-fix findings, descriptive-field checks, fixed-field alignment, subjects, classification, local cleanup, and acceptable-as-is elements. Third, the cataloger validated the output against the physical item, Connexion or Alma, authority data, standards, and local practices. Fourth, each finding was either accepted, modified, rejected, or recorded as a missed issue or unsupported assumption. Fifth, any correction with broader applicability was incorporated as a specification change and tested in subsequent cases.

The research workbook comprised a record review log, a finding-level log, a specification-change log, an error taxonomy, and a dashboard. An accompanying human-readable specification organized the review process into recurring sections, including descriptive fields, fixed-field alignment, subjects, classification, local cleanup, acceptable-as-is elements, and research notes.

As of this revision, the human-readable checklist included 92 operational specifications, and the evolution log recorded 24 changes. The most detailed sections addressed the 245 title statement field, general review principles, call-number review, 264 publication/copyright statement fields, physical description, notes, and subject access. The checklist's extensive length emerged as a significant finding, illustrating how seemingly straightforward cataloging assistance evolved into a complex specification when legacy practices, copy-and-paste artifacts, item evidence, Special Collections considerations, and local policies were systematically addressed.

Since row-level outcome coding is complete only for selected examples and the extended-evidence case, it is premature to estimate overall precision, recall, false-positive rates, or time savings. The findings presented here are limited to qualitative patterns and the documented evolution of specifications.

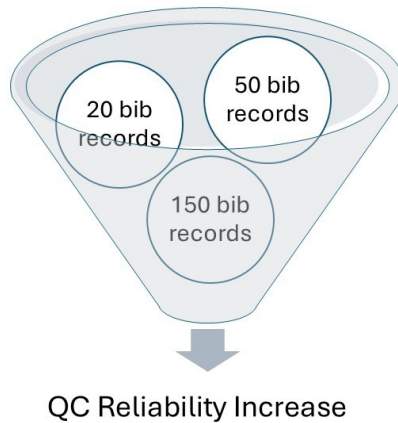


Figure 1.2. Catalog Buddy GPT development: Funnel illustrates the number of bibliographic records reviewed, with Cataloging Buddy GPT evaluating additional quality control specifications at each stage to provide increasingly detailed metadata assessment.

Hell or high water serves as the most clearly coded example of the division of labor. The assistant detected recurring punctuation issues in the 245 title statement field, 260 AACR2 publication field, and 300 physical description field. The cataloger accepted these corrections, added a date to the call number in accordance with local practice, determined that a MacArthur military-leadership subject heading was justified by the subtitle and content, and required verification of a pasted 505 contents note field display artifact. This record thus demonstrated both accepted machine-generated findings and cataloger-only decisions.

Other cases contributed to specification changes in various ways. *A slow life but a good life* demonstrated that item-in-hand evidence takes precedence over related-work dates or vendor inferences. *Clemson University campus planning & design, 2000–2020* established that subject headings and classification must reflect the same conceptual focus, and that personal, corporate, and local Cutter assignments cannot be inferred solely from topical terms. *Violent Femmes* clarified the distinction between a series statement and the title proper, discouraging the creation of artificial controlled access for album-focused records. *Textile wet processes* established manuscript rules for the 264 _0 production statement field, leaves, form notes, and relationships to subsequent publications. *Thunder of the captains* and *The day the Chinese attacked* illustrated that systems should be recognized for restraint when a legacy record is largely usable.

The evolution log indicates that cataloger disagreements contributed significantly to the refinement of the specifications. New rules addressed RDA and AACR2 contexts, 008 fixed-length coded data field workflow, 245 title statement field transcription accuracy, geographic area codes, biography thresholds, 700 personal-name added entry field and 710 corporate-name added entry field indicator positions, manifestation-specific ISBNs, inherited volume numbering, local holdings notation, and the distinction between research data and production

output. The assistant's performance improved as the specification more clearly delineated the boundaries of cataloger judgment.

4.9 Catalog Buddy: limitations and next phase

The 20-record corpus is limited in size, purposively selected, and derived from a single cataloger's workflow. As a result, it disproportionately reflects academic and Special Collections priorities and does not support generalizations across all libraries, languages, formats, or catalogers. Additionally, pasted MARC data may collapse spaces, indicators, and tags, resulting in display artifacts that do not constitute errors in the editable record. The assistant also lacked consistent access to current authority files, Classification Web, and local system configurations. Furthermore, incomplete outcome coding precludes reliable estimates of accepted findings, false positives, missed errors, or time savings.

The subsequent phase should involve finalizing a specification version, completing row-level coding for the development corpus, and evaluating a separate set of records not used in rule creation. At least two catalogers should independently assess a subset to measure inter-reviewer agreement. Relevant metrics include field-level precision and recall, severe false negatives, time spent reviewing false positives, accepted suggestions per record, differences by record type, and the proportion of findings dependent on item evidence, authority research, or local policy. Any model update should prompt retesting rather than unmonitored continuation.

5. Quality Control and Data Integrity

5.1 Why quality control is a practical starting point

Metadata quality should be understood as an ongoing process rather than a single checkpoint. Yang et al. (2025) outline a lifecycle comprising acquisition, cleaning, verification, and maintenance, with each stage contributing to accuracy and long-term utility. AI-assisted quality control is therefore more effective as a component of continuous improvement rather than as a one-time remediation effort.

Quality control serves as a practical entry point because automated systems can be restricted to identifying records that require further attention, without determining the final content of the record. Proffitt and Dortmund (2025) highlight backlogs, scripts, and languages lacking local expertise, as well as metadata cleanup, as domains where AI can provide support. Their recommendation is pragmatic: expand capacity while maintaining human expertise as central to review and guidance. This division of labor—machine flagging followed by cataloger review—is also reflected in the PCC environmental scan (PCC Task Group 2024).

Quality control encompasses a range of issues, including date and language-code conflicts, duplicate records, unsupported subjects, incorrect indicators, missing subfields, and fixed-field mismatches. Pattern-based tools facilitate the efficient scanning of large batches and enable comparison of record versions during migration or remediation.

Performance is contingent upon local configuration. The review framework should differentiate between mandatory corrections, optional normalization, accepted local practices, and

interpretive questions. Without these categories, automated tools may generate as much irrelevant output as they resolve.

5.2 Structural errors, duplicates, and local practice

At scale, AI and conventional validation methods can be integrated. Deterministic rules are suitable when standards are explicit, such as when a required subfield is missing, an indicator falls outside its valid range, or a fixed-field value conflicts with a coded field. In contrast, statistical or language-based methods are advantageous for contextual problems, such as identifying potential duplicates with variant titles, detecting notes that paraphrase title fields, or evaluating uncertain authority matches.

Duplicate detection exemplifies a layered review process. Automated systems can compare identifiers, normalized titles, creators, dates, editions, physical descriptions, and holdings to group likely duplicates and rank them by confidence. Catalogers then determine whether records represent the same manifestation, related editions, reproductions, or distinct resources. Automatic merging should be limited to cases governed by robust, testable rules and supported by a reversible audit trail.

Validation processes must differentiate between genuine errors and acceptable variations in practice. For example, records created under the Anglo-American Cataloging Rules, 2nd edition (AACR2), the use of an older **260 AACR2 publication field**, retention of vendor data, missing final punctuation, or a blank local call-number field may be contextually appropriate. Automated checkers that flag every deviation from current preferences as a defect risk, generating excessive noise and potentially erasing valuable record history.

Corrado's (2021) principle of balancing metadata quality with return on investment is equally relevant to remediation: not every legacy variation warrants recataloging. Tools that fail to distinguish between harmful errors and acceptable historical practices may ultimately require more staff time than they save.

5.3 Field-level review, anomaly detection, and data integrity

A practical AI-assisted QC workflow can be organized around specific MARC areas. For the title statement, the tool may check **245 title statement field** indicators, nonfiling characters, the use of subfield \$b for true other title information, and punctuation before the statement of responsibility. For production, publication, distribution, manufacture, and copyright fields, the tool may check whether **264 production/publication/copyright statement field** functions match the evidence and whether fixed-field date coding aligns with publication and copyright dates. For the physical description, the tool may compare the **300 physical description field** with the item in hand and check whether illustrations, maps, plans, color content, or leaves are described correctly.

Notes and summaries also require careful review. A **500 general note field** may document the source of the title, item format, limited edition information, or report context. A **520 summary note field** should remain neutral and avoid promotional language from the publisher. A **504 bibliography/index note field** should be included only when the item contains bibliographical

references or an index, as this evidence influences fixed-field coding. While AI tools can prompt catalogers to review these areas, ultimate verification remains the cataloger's responsibility.

Access points and fixed fields similarly demand careful attention. Added entries must include correct indicators, relationship designators, punctuation, and authority forms. The **043 geographic area code field** should correspond to the item's geographic coverage and the geographic subdivisions in subject fields. The **008 fixed-length coded data field** should be constructed or reviewed only after variable fields are stabilized, particularly those related to dates, physical description, bibliography, or index evidence, and subject coverage. In these areas, AI can assist by prompting review, but should not replace evidence-based cataloging.

Anomaly detection can prioritize records that deviate from expected patterns, such as unusual combinations of **008 fixed-length coded data field** values and variable fields, unexpected date sequences, rare or missing fields for a given resource type, geographic codes misaligned with subject headings, or abrupt changes in vendor data. Predictive analytics can further estimate which records are most likely to require significant remediation, thereby assisting managers in constructing review queues and allocating staff resources.

These methods are intended to predict review priority rather than establish bibliographic accuracy. An atypical record may be correct, while a common pattern could be systematically erroneous. Evaluation metrics should include false positives, false negatives, precision at the top of the review queue, time saved, reviewer agreement, and the types of errors the system fails to detect. Recalibration is necessary when collections, vendors, standards, or local policies are updated.

Transparent quality control outputs should specify what was checked, the evidence available, the reason for flagging an item, the system's level of certainty, and the recommended human action. A concise overall status—such as ready for review, minor edits, moderate cleanup, major work, or original cataloging required—can assist staff in prioritizing tasks, provided that the underlying rationale remains accessible.

5.4 Byrnes photograph metadata project: purpose and method

The second project addresses a more focused collection-level question: can a rule-guided assistant triage photograph metadata for title style, description presence or duplication, and date consistency without interpreting image content? The source workbook for the James F. Byrnes photograph collection comprises 2,400 raw metadata records, a comprehensive clean-data worksheet, and project notes. The experimental file isolates clean-data rows 1442–1921, representing a contiguous sample of 480 records, or 20.0% of the dataset, spanning identifiers Mss90_1466 through Mss90_1945.

The unit of analysis is a single metadata row, rather than a photograph as visually interpreted by a model. The review utilized the existing Title and Date, Created, Approximate.Date, Description, Place, State, Country, and Subjects/Names fields. The process did not involve inspecting image pixels, identifying individuals, verifying historical claims, or validating names and subjects against authority files. Consequently, a “Pass” indicates compliance with the specified metadata rules, but does not guarantee that the description is complete, unbiased, or visually verified.

The quality control rules were deliberately conservative, focusing on repeatable title and date issues without generating new descriptive content. Blank descriptions were allowed, as descriptions were optional during this phase. Replacement titles were suggested only when capitalization or tense could be corrected mechanically; ambiguous cases were referred for further review.

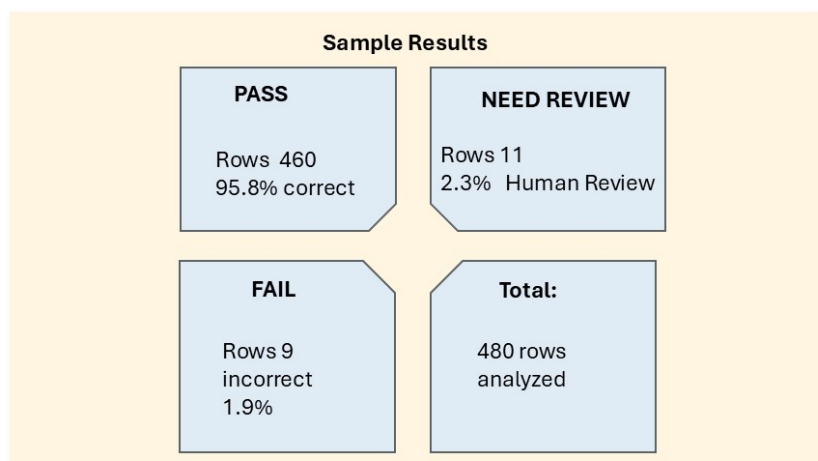
For each metadata row, the workflow assigned separate Title QC, Description QC, and Date QC values, followed by an Overall Status of Pass, Review, or Fail. 'Fail' was reserved for clear rule violations with mechanical corrections, such as sentence-style capitalization or present-progressive wording. 'Review' was applied to vague titles and to date conflicts that required contextual assessment. The Needs Work sheet compiled Review and Fail rows, while the QC Summary tabulated statuses and issue types. Confidence was rated as High for mechanical or no-issue outcomes and Medium for cases requiring contextual review. Notably, the workflow could be piloted prior to a comprehensive quality-control review. If at least 85 percent of records passed the defined checks, the collection was considered lower risk for those specific issues, though further review remained necessary for areas outside the test's scope.

5.5 Byrnes Photograph Metadata Project: results and limitations

The initial quality control summary classified 460 of 480 records as Pass, 11 as Review, and 9 as Fail. Thus, 95.8% of records passed the initial screen, while 4.2% required further attention. These figures represent preliminary screening results rather than post-correction accuracy measures.

The flagged records encompassed six primary issue types: 9 vague titles, 7 capitalization errors, 2 title-tense issues, 1 duplicated or mixed date phrase, and 1 title/date year conflict. The workflow proposed nine replacement titles—corresponding to the seven capitalization and two tense corrections—since these were mechanical repairs. The eleven contextual cases were reserved for human review rather than resolved through inference.

Figure 1.3 Sample: summarizes the final Byrnes metadata screen for a post-QC sample of Byrnes records. The high pass rate reflects both the limited scope of the rules evaluated and the effects of prior quality control, rather than indicating complete metadata accuracy.



The working workbook was used iteratively, with some flagged titles corrected after the initial screening, while the QC Summary retained the original counts. As a result, the current row-level states and static summary do not represent a controlled pretest/posttest dataset. Reporting only the initial screen does not equate to a claim of final correction accuracy for an evolving work file.

The results demonstrate that a narrowly defined rule set can reduce a 480-record review queue to 20 targeted records. However, this does not indicate that 95.8% of the metadata is fully accurate. The screening process did not compare titles and descriptions with images, assess the accuracy of names and subjects, evaluate inclusive language, or determine whether blank descriptions warranted enrichment. Additionally, repeated records may inflate the pass rate when a single title pattern recurs. The contiguous sample was selected for convenience and boundedness, rather than through random or stratified sampling, and may not be representative of the entire Byrnes collection.

The status labels reflect underlying policy decisions. Accepting blank descriptions is appropriate for a title-focused experiment but would not be suitable for a project aiming for comprehensive descriptive enrichment. Similarly, sentence-style capitalization and present-progressive wording are local quality standards rather than universal principles for archival description. Consequently, the results must be interpreted in the context of the specific rule set and scope that generated them.

Photograph collections introduce ethical considerations that cannot be addressed solely through spreadsheet analysis. Issues such as names, relationships, cultural contexts, historical claims, harmful legacy language, and local significance may necessitate visual inspection, archival context, authority research, and consultation with the communities represented in the collection. The assistant can direct these questions for further review, but should not resolve them autonomously.

The subsequent phase should lock the source sample, rule version, and initial outputs prior to any corrections. A stratified or random validation sample should encompass various date ranges, locations, event types, levels of description, and records both with and without descriptions. Catalogers should examine the corresponding images and independently assess a subset. Post-correction audits should be maintained separately from the initial screen to enable measurement of improvement. Additional evaluation metrics should include false positives, undetected visual or authority errors, reviewer agreement, correction time, recurring issue clusters, and the impact of rule changes on legacy records.

Only after completing this validation should the rules be applied to the entire set of 2,400 records. Even at that stage, collection-wide automation should remain reversible: the source value, generated suggestion, reviewer, decision date, and final value should all be preserved to ensure that any erroneous correction can be traced and reversed.

5.6 Comparative lessons from the local projects

Although the projects utilize different types of evidence, both support a common 'buddy' model. This model is characterized by a narrowly defined machine role, transparent evidence, documented human decisions, and mechanisms for carrying corrections forward into subsequent reviews.

The buddy model represents a workflow rather than a specific product. Project 1 requires both field-level and decision-level validation, while Project 2 necessitates a distinct post-correction audit.

6. Ethical and Practical Considerations

6.1 Fluent error and automation bias

One difficulty with language models is that an error can arrive in a finished-sounding sentence. In a catalog record, that may mean an invented publication statement, a nonexistent subject heading, an unsupported relationship, or a confident explanation of a rule that does not apply. These errors become harder to catch when reviewers are rushed or unfamiliar with the material.

Metadata errors also travel. A false authority match can attach a work to the wrong person; a misleading subject can distort retrieval; an unsupported summary can misrepresent the resource. Once records are shared, migrated, harvested, or reused as linked data, even a small mistake may persist across systems.

Concern about fluent error is not confined to cataloging experiments. Cher (2025) notes that ChatGPT-based services can deviate from factual accuracy and reproduce social bias, while Roy et al. (2024) identify privacy, algorithmic bias, and human oversight as continuing concerns in automated cataloging and classification. These risks are operational rather than abstract because a false or biased description can become part of the public record.

Automation bias compounds the problem. Reviewers may defer to polished wording or a confidence score, especially when checking takes longer than generation. High-risk fields should therefore present supporting evidence and receive independent verification; periodic blind comparisons with human-created work can reveal whether the review process has become too trusting.

The local projects illustrate a second form of automation bias: a well-structured output or high pass rate can look more comprehensive than it is. Catalog Buddy could sound certain about unsupported dates or local treatment, and the Byrnes 95.8% pass rate applied only to a narrow title-, description-, and date-focused screen. Reviewers need the task boundary and denominator in view whenever they interpret a result.

6.2 Authority gaps, bias, and local context

Many AI tools do not consult live authority files or current cataloging documentation unless the workflow explicitly provides access. Suggestions must still be checked against LCSH, the Library of Congress Name Authority File (LCNAF), FAST, Classification Web, RDA, MARC documentation, OCLC Bibliographic Formats and Standards, and local policy as appropriate.

Bias enters from more than one source. Models reproduce patterns in training data, while catalogs and controlled vocabularies carry their own histories of exclusion and harmful language. AI-assisted description can combine both sets of problems unless someone examines whose language is being repeated and whose context is missing.

Local context is another blind spot. A generic system will not know why a field is retained, why a legacy record is considered sufficient, how an institutional Cutter works, or what relationship a library has with a represented community unless that information is supplied.

An authorized term can still be too broad, misleading, or harmful in context. The review must also ask whether access is fair, specific, contextualized, and consistent with an institution's inclusive description policy (Dover and Grzegorski 2025).

Libraries also occupy a distinctive ethical position. Dover and Grzegorski (2025) remind readers that “the public has traditionally placed its trust” in libraries to help navigate information (p. 603), yet profession-wide guidance on responsible AI use in cataloging remains underdeveloped. They advocate a cataloging-specific AI code of ethics and a practical checklist that staff could use at the point of decision (Dover and Grzegorski 2025, p. 614).

6.3 Transparency, privacy, and data governance

Transparency should match the use. Internally, staff need a record of the tool and model, the data supplied, the prompt or template, the accepted changes, and the reviewer. Public disclosure may be appropriate at the collection, project, or record level when AI materially shapes description. The published photograph projects show one way to make that involvement visible.

Operational explainability is usually more useful than a technical account of model internals. Staff and users need to know which task was automated, which evidence was used, where uncertainty remains, who approved the result, and how an error can be reported and corrected.

Sensitive metadata, donor information, internal notes, unpublished descriptions, and patron-related data should not be entered into a third-party service without institutional approval. Before adoption, the library needs clear answers about prompt storage, model training, access, retention, and contractual protections.

Kalbande et al. (2025) similarly connect AI adoption with privacy risk, algorithmic bias, and changing professional roles. Roy et al. (2024) argue that ongoing research and explicit ethical attention are necessary if libraries are to benefit from these systems without their risks becoming routine.

Vendor-embedded AI creates a different problem: staff may receive recommendations without visibility into the data, model, or error rate behind them. AI literacy, therefore, matters even for employees who never open a standalone chatbot.

Data-governance rules should address prompt retention, training use, access controls, storage location, security review, copyright, and restricted or culturally sensitive collections. Convenience is not a privacy assessment.

6.4 Staff roles, labor, and professional development

Labor claims deserve the same scrutiny as technical claims. Automating a repetitive step may reduce one burden while creating others: output review, error documentation, bias assessment, prompt maintenance, training, and policy work. Those tasks require expertise and time.

Chen and Li's (2024) respondents saw potential for record analysis and creation but also worried about the further deprofessionalization of cataloging. That concern should be treated as a workflow-design issue. If staff are reduced to correcting opaque output, the technology may erode both job quality and the expertise needed to detect subtle errors.

Moving staff from description to continuous correction can make the work monotonous and weaken foundational skills. A responsible workflow preserves meaningful cataloging, provides time for training and documentation, and recognizes review as skilled labor rather than invisible cleanup.

Environmental cost belongs in procurement and project assessment alongside price and accuracy. The PCC survey found that ecological concerns appeared less often than misinformation or bias, but they were still present. Libraries can limit unnecessary regeneration, reuse tested prompts, and choose the least resource-intensive tool that meets the task (PCC Task Group 2024).

7. Implementation Strategies

7.1 Define a bounded, low-risk pilot

Implementation begins with a testable question. 'Can AI do cataloging?' is too broad; 'Can this tool identify **245 title statement field** indicator conflicts in a defined record set?' or 'Can it triage title and date inconsistencies in a frozen photograph-metadata sample?' can be evaluated. Professional guidance favors bounded tasks, expert involvement, and realistic expectations over generalized cost-saving claims (PCC Task Group 2024).

Early pilots should use low-risk, non-sensitive data and a review set with known answers or well-documented human decisions. The pilot plan should name the sample, expected output, review checklist, error categories, documentation method, responsible staff, and decision point. It should also assign risk before testing: formatting public text or flagging deterministic conflicts is lower risk, while identity resolution, sensitive description, rights decisions, and automatic record replacement should be treated as high risk.

Proffitt et al. (2024) recommend a deliberately paced, collaborative approach built around domain-specific data, high-quality metadata, and safe places for staff learning. A pilot should not become production simply because early output looks promising; it should move forward only when evidence, risk, and review cost have been documented.

7.2 Build prompts, templates, logs, and stop rules around evidence

Good prompting begins with cataloging context, not clever wording. Instructions should state the task, resource type, available evidence, relevant standard, local rule, prohibited behavior, and expected output. Prompting is not a substitute for expertise; Iacono's (2025d) phrase "Teach judgement, not prompts" captures the need to train staff to evaluate answers rather than merely elicit them.

Reusable templates make outputs comparable. A quality-control template can separate must-fix errors, optional normalization, local practice, and judgment calls; a summary template can set

length, tone, and evidence limits. Logs should record the tool and model, data supplied, task, output, reviewer, accepted changes, rejected suggestions, and observed errors. Prompts should also state when the system must stop, such as “Do not supply missing data,” “mark unverified authority candidates,” or “require item-in-hand review.”

7.3 Measure performance, harm, and review burden

Metrics should match the task. Useful measures may include field-level accuracy, precision and recall of error flags, review time, severe false negatives, reviewer agreement, and the share of output requiring major revision. Initial screening, correction, and post-correction audit results should be stored separately so that early triage is not mistaken for final accuracy.

Average accuracy alone is a poor go/no-go rule. A system may appear efficient while producing rare but damaging errors. Stop conditions should be set in advance, including privacy risk, unverifiable output, repeated identity mistakes, unmitigated bias, excessive review burden, unstable behavior, or loss of necessary vendor controls. A well-documented decision to stop is a successful pilot outcome when it prevents harm.

7.4 Train staff and move into governed workflow

Training should cover how models produce output, why confident answers can be wrong, what information may not be shared, and where staff should verify headings, names, classifications, fields, indicators, and local policy. Harisanty et al. (2024) emphasize competencies such as analytical and critical thinking, adaptability, communication, creativity, and leadership. Dover and Grzegorski (2025, p. 601) note that the 2023 revision of the Core Competencies for Cataloging and Metadata Professional Librarians added understanding AI’s impact, limitations, and possible uses as a knowledge competency.

AI literacy also involves understanding where AI is already reshaping routine professional work. Iacono (2025a) frames AI as a quiet workplace shift rather than a replacement, while Iacono (2025b) argues that human-AI partnership requires staff to upgrade familiar professional skills rather than abandon them.

A responsible implementation cycle should define the task and evidence, design the workflow, test a representative sample, evaluate value and harm, make a documented go/no-go decision, operate with oversight, and reassess periodically. Each stage should leave a record that another person can review later. Treating AI as metadata governance avoids the assumption that every pilot should move toward autonomy.

Implementation also requires partnerships. Affum and Dwomoh (2023) emphasize collaboration among libraries, researchers, and AI specialists, while Bridges et al. (2024) show how partnerships can amplify library perspectives in institutional and governmental discussions. Large-scale initiatives, such as the Boston Public Library’s 2025 AI-assisted digitization announcement, should be read as evidence of institutional direction rather than proof of completed success (Boston Public Library 2025). Success is measured not by removing people from the process but by placing human attention where evidence, ethics, and accountability matter most.

8. Conclusion

AI is already entering technical services work through extraction, prediction, generation, and quality-control tools. Some uses are practical: drafting from supplied evidence, comparing records, ranking candidates, finding anomalies, and extending review across large collections. The two local projects make the boundary visible. Catalog Buddy turned 20 MARC cases into an evolving specification and qualitative evidence about structural checks, local practice, and cataloger judgment. The Byrnes project reduced a 480-record title-focused screen to 20 records needing attention, but it did not validate the photographs or establish a post-correction accuracy rate. Neither project supports autonomous cataloging.

What matters now is whether a library can define what was tested, verify the result, explain the workflow, and absorb the cost of review. Cataloging still depends on evidence, standards, local context, ethical judgment, and responsibility for discovery. Those requirements become more important when output can be generated or screened at scale.

A useful partnership keeps the machine's role narrow and the cataloger's authority clear. The assistant may notice patterns, draft mechanical corrections, or build a review queue. Trained staff decide what the resource is, how it should be represented, whether the description is fair and useful, and whether a change belongs in the public record. Cataloger disagreements and corrections are not obstacles to remove; they are evidence that improves the workflow.

Progress should be judged by record quality, discoverability, transparency, equitable description, and the working conditions of the staff who maintain the data. That standard is consistent with Ranganathan's user-centered laws: resources should be usable and discoverable, readers should find what they need without unnecessary delay, and the library should continue to grow with its community (Kalbande et al. 2025). AI-supported metadata can improve discovery and reuse only when acquisition, cleaning, verification, and maintenance are treated as a single continuous lifecycle (Yang et al. 2025). Professional judgment must remain visible and accountable throughout that work.

Figure captions

Figure 1.1. Decision categories for AI-assisted cataloging quality control. The figure summarizes five review categories used to distinguish must-fix errors from high-risk review issues, optional normalization, accepted local practice, and records that may be accepted as-is.

Figure 1.2. Catalog Buddy development outputs. The Catalog Buddy development corpus included 20 MARC records and produced 92 operational specifications and 24 documented evolution-log changes. These figures describe specification growth and workflow complexity, not final precision, recall, or time savings.

Figure 1.3. Byrnes photograph metadata QC screening outcomes. The initial screen classified 460 of 480 records as Pass, 11 as Review, and 9 as Fail. The 20 flagged records included vague titles, capitalization errors, title-tense issues, a duplicated or mixed date phrase, and a title/date year conflict. These results represent a limited rule-guided screen rather than full descriptive or visual verification.

Table 1.2. Glossary Key terms used in this chapter

Term	Plain-language definition	Use in this chapter
Artificial intelligence (AI)	Computer systems that identify patterns, generate outputs, compare information, or support decision-making.	Used as the broad category for tools that may assist metadata, cataloging, and quality-control work.
Generative AI	AI that creates or transforms text, summaries, code, images, or other content based on patterns in data.	Discussed as a tool for drafting summaries, suggesting metadata, and supporting review.
Large language model (LLM)	A type of generative AI designed to process and produce language-based output.	Used for language-based tools such as chatbots or configurable assistants that generate text.
Machine learning (ML)	A form of AI that identifies patterns in data and uses those patterns to make predictions or classifications.	Used in discussions of authority matching, duplicate detection, classification support, and anomaly detection.
Technical services	Library work involved in acquiring, organizing, describing, managing, and maintaining access to resources.	Serves as the main professional setting for the chapter.
Cataloging	The process of creating and maintaining descriptions that help users find, identify, select, and access resources.	Used to discuss MARC records, classification, subject access, and authority work.
Metadata	Structured information that describes, identifies, manages, or supports discovery of resources.	Used for catalog records, digital collections, repository description, and photograph metadata.
Metadata quality	The accuracy, consistency, completeness, clarity, and usefulness of metadata.	Used to evaluate whether AI-assisted work improves discovery or creates new review burdens.

Term	Plain-language definition	Use in this chapter
Cataloger judgment	Professional reasoning used to interpret evidence, apply standards, and make cataloging decisions.	The central concept of the chapter; AI can assist, but catalogers decide.
Human-in-the-loop	A workflow in which a person reviews, verifies, and approves output before it is accepted.	Used to describe responsible AI-assisted cataloging and metadata review.
AI literacy	The ability to understand, evaluate, and responsibly use AI tools and outputs.	Treated as a professional competency for technical services staff.
Responsible AI	The use of AI with attention to fairness, privacy, transparency, accountability, and human oversight.	Used to frame ethical implementation in library workflows.
MARC	A structured format that uses coded fields to store and exchange bibliographic information.	Used in discussions of record review, field-level quality control, and Catalog Buddy.
BIBFRAME	A linked-data model intended to describe bibliographic resources through entities and relationships.	Discussed as part of linked-data and MARC-to-BIBFRAME workflow possibilities.
RDA	Resource Description and Access, a cataloging standard used to describe library resources.	Used when discussing standards-based cataloging decisions and descriptive consistency.
Authority control	The process of using consistent, authorized forms of names, subjects, titles, or entities.	Used in discussions of name disambiguation, subject verification, and linked-data reliability.
Subject analysis	The process of determining what a resource is primarily about.	Used to show why AI suggestions require human judgment and review.

Term	Plain-language definition	Use in this chapter
Controlled vocabulary	A standardized list of approved terms used to describe subjects, genres, names, or concepts consistently.	Used when discussing LCSH, FAST, and AI-suggested subject terms.
Quality control (QC)	Review work that checks records or metadata for errors, inconsistencies, omissions, or unsupported claims.	Used as one of the chapter's strongest practical areas for AI assistance.
Metadata remediation	The process of improving, correcting, standardizing, or enhancing existing metadata.	Used for cleanup projects, legacy records, and batch metadata review.

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